Western Hognose Snake Care Guide

Heterodon nasicus

Provide deep diggable substrate, several hides and a reliable whole-prey feeding routine. Defensive hissing and bluff strikes are normal possibilities; repeated disturbance and oversized meals can worsen feeding problems.

Species Profile

Common name: Western Hognose Snake

Scientific name: *Heterodon nasicus*

Adult size: Males commonly 14-24 in (36-61 cm), females 24-36 in (61-91 cm) total length

Average lifespan: 15-20 years

Native range: Central North America, southern Canada through the United States to northern Mexico

Habitat and activity: Diurnal, terrestrial and burrowing; prairie/sandy habitats

Diet: Appropriately sized frozen-thawed whole mice for established captive animals

Care difficulty: Intermediate

Shopping List

- Secure enclosure at least 36 × 18 × 18 in (91 × 46 × 46 cm; L × W × H) for one adult; see Housing for exceptions and group arrangements.
- Digital thermometers for different zones; a suitable hygrometer; timer for a consistent day/night cycle.
- Species-appropriate substrate and multiple stable hides or covered retreats.
- Reliable supply of the staple foods described below, clean feeding tools and a transport container.
- Appropriate overhead heat source if required, controlled by a compatible thermostat; infrared thermometer for surfaces.
- Appropriate visible lighting and, where recommended below, a linear UVB fixture with measured shade and exposure zones.
- Fresh-water dish or cup; insect supplements when applicable; a scale appropriate to the animal's size.

Housing

Our recommended starting adult enclosure is 36 × 18 × 18 in (91 × 46 × 46 cm; L × W × H) for one animal. Larger usable space is encouraged. These dimensions are a husbandry recommendation, not a species-specific experimentally proven threshold. Use at least 48 × 24 × 24 in (122 × 61 × 61 cm) for a large adult female or any snake unable to stretch fully in the smaller enclosure.

Provide multiple hides across the warm-to-cool and moist-to-drier gradient, covered travel routes and species-appropriate burrowing or low climbing structure. Fix heavy decor to the floor before adding loose substrate so digging cannot undermine it.

Juveniles may use a furnished adult enclosure when food intake, hydration and waste can be monitored. A temporary grow-out enclosure should still offer choice and cover. Quarantine newcomers separately with dedicated tools and a substrate that permits health monitoring; arrange veterinary assessment for unknown-origin animals.

Lighting & UVB

Provide roughly 10-12 hours of visible daylight followed by darkness. A measured UV Index (UVI) around 0.5-1.0 at the most exposed usable site, declining to near zero inside retreats, is our conservative starting recommendation. Low-level UVB is recommended when a usable gradient can be provided; nocturnal activity alone is not a reason to exclude light.

Measure exposure at the animal's back height with an appropriate UVI meter, such as a Solarmeter 6.5. Lamp percentage, wattage and a fixed distance alone do not establish exposure: reflectors, mesh, perch height, plant growth and lamp age change output. Keep glass/plastic out of the UV path, follow fixture safety instructions and provide complete shade. Replace or reassess lamps according to measured output and manufacturer guidance.

Heating & Temperature

- Localized basking surface: 90-95°F (32-35°C).
- Cool-zone air: 70-77°F (21-25°C).
- Nighttime air: 65-75°F (18-24°C) for healthy established animals.

Create the gradient on one side rather than heating every hide to the same temperature. Use a compatible thermostat on each heat source; a manual lamp dimmer does not replace thermostatic protection. Secure the probe and verify temperatures independently: probe thermometers measure air/retreat temperature, while an infrared thermometer measures exposed surfaces.

Use appropriately guarded overhead warmth when needed. Keep animals clear of hot fixtures, avoid heat rocks, and switch visible lights off at night. Night heat is needed only when the room falls below the appropriate range; use a thermostat-controlled lightless source if necessary.

Humidity & Ventilation

30-50% with a humid retreat available. Treat readings as a starting pattern rather than a number to chase regardless of substrate, airflow or animal behavior.

Measure humidity in a representative occupied zone, away from a freshly sprayed surface. Provide sheltered moisture choices while retaining airflow. Persistent condensation, sour odor, flooded soil or surfaces that never dry indicate a setup problem. Ventilation and hydration must be adjusted together rather than simply sealing all vents.

Water & Hydration

Provide a clean, appropriately shallow dish or accessible elevated water cup at all times. Change it daily and immediately when soiled; wash it regularly with dedicated tools. Potable water is generally suitable, with treatment where local water quality requires it.

For animals that drink droplets, mist foliage or suitable surfaces at the time they become active, then allow partial drying. Droplets supplement the dish; water chemistry claims should be based on evidence rather than a species' trade name.

Substrate

Substrate is extremely important for western hognose snakes because burrowing is one of their main natural behaviors. Provide at least 3-4" of substrate for juveniles and 4-6" for adults whenever possible.

Aspen shavings or aspen chips can work well for a simple setup as long as they are dry, clean, and not dusty. For a more naturalistic setup, a useful DIY mix is around 70% organic topsoil and 30% washed play sand, with leaf litter and cork added on top. Avoid cedar, pine, scented bedding, dusty sand, and wet substrate.

Feces and urates should be removed as noticed, and soiled substrate should be scooped out and replaced. Full substrate changes are usually done every 3-6 months depending on enclosure size, cleanliness, and the type of substrate used.

Food & Feeding

Western hognose snakes are carnivores that eat whole prey. In the wild they are well known for eating toads and other small animals, but captive-bred hognoses should be maintained primarily on frozen/thawed mice. Do not feed wild amphibians, wild rodents, or wild-caught prey because of parasite, pesticide, and toxin risk.

Prey should be appropriately sized, usually about the same width as the thickest part of the snake or around 10-15% of the snake's body weight. Frozen prey should be fully thawed and warmed before offering. Live prey is not recommended because it can injure the snake.

How often western hognose snakes need to eat depends on age, sex, and body condition:

Hatchlings and young juveniles - 1 appropriately sized pinky every 4-5 days

Juveniles and subadults - 1 appropriately sized mouse every 5-7 days

Adults - 1 appropriately sized mouse every 7-14 days

Adult females often eat more reliably and may need meals closer to every 7-10 days, while adult males may maintain better condition on a 10-14 day schedule. Avoid overfeeding, as hognose snakes can become overweight in captivity.

Supplements

A varied, appropriate whole-rodent feeding program does not normally require calcium or vitamin dusting. Do not supplement whole prey routinely unless a reptile-experienced veterinarian identifies a specific need.

Decor & Enrichment

Decorations help western hognose snakes feel secure and encourage natural behavior. Good decor can include cork bark, flat hides, half logs, rocks, artificial plants, leaf litter, low branches, and clumps of substrate that allow the snake to burrow and explore.

Provide at least two snug hides: one on the warm side and one on the cool side. More cover usually results in a calmer snake that is more likely to explore during the day. Open, bare enclosures can make hognoses feel exposed and may lead to defensive behavior or feeding issues.

Handling & Behavior

Let the animal settle and feed consistently before brief optional handling. Support the body, work low over a safe surface and stop when repeated retreat, escape attempts or defensive behavior show stress. Support the snake’s whole body and never lift it by the tail.

Avoid handling immediately after meals, during difficult sheds or when unwell. Snakes should usually be left undisturbed for at least 48 hours after feeding. Wash hands and use dedicated equipment to reduce disease transfer.

Hognoses have enlarged rear teeth and biologically active saliva. A prolonged feeding bite can cause significant local swelling; never encourage chewing, handle around food odors or dismiss a bite reaction. Seek medical help for severe or spreading symptoms.

Cleaning & Routine Maintenance

Daily: check the animal's access to water/appropriate moisture, temperatures, visible condition and equipment. Remove feces, contaminated material, dead feeders and spoiled food promptly.

Regularly: clean dishes and feeding surfaces, check thermostats and probes, inspect decor stability and maintain ventilation. Bioactive systems still require waste checks, food removal, plant care and drainage maintenance; they are not self-cleaning.

Record appetite, activity, shed history and weight or body condition when safe. Arrange veterinary assessment for unexplained changes. Quarantine new animals with dedicated tools, and disinfect only with a suitable product used at the label dilution/contact time, then rinse and dry as required before return.

Cohabitation

Individual housing is the safest routine default. Shared housing can cause food competition, injury or reproductive pressure even without obvious fighting. Never assume a male/female pair or female group is compatible. Breeding introductions belong in a planned specialist program with separate backup enclosures and individual monitoring. Size differences and predatory behavior can lead to cannibalism; never feed animals together.

Shedding & Seasonal Changes

Maintain access to a clean humid retreat and inspect toes, tail tip and eyes after shedding when observation is possible. Do not peel retained skin or apply oils/adhesives. Persistent constriction, damaged tissue or repeated failed sheds warrants veterinary help.

Seasonal changes in appetite or activity can occur. Check temperatures, hydration and weight before treating prolonged refusal as normal. Deliberate brumation or breeding cycles require healthy animals and species-specific preparation; pet care does not require forcing a seasonal shutdown.

Common Health Concerns

- Poor body condition or feeding decline: check the full gradient, hydration and food quality; arrange assessment for persistent refusal accompanied by weight loss.
- Calcium/vitamin imbalance: weakness, tremors, a soft or misshapen jaw or abnormal limbs require veterinary assessment; extra high-dose supplements are not a substitute for diagnosis.
- Skin, shed or injury problems: repeated retained skin, wounds, swelling or abnormal discoloration deserve prompt attention.

- Respiratory or systemic illness: labored/open-mouth breathing at rest, marked lethargy, loss of balance, persistent diarrhea or rapid deterioration require an experienced exotic veterinarian.